

Clinical Epidemiology of Prostate Cancer

Mieke Van Hemelrijck, PhD
Lecturer in Cancer Epidemiology

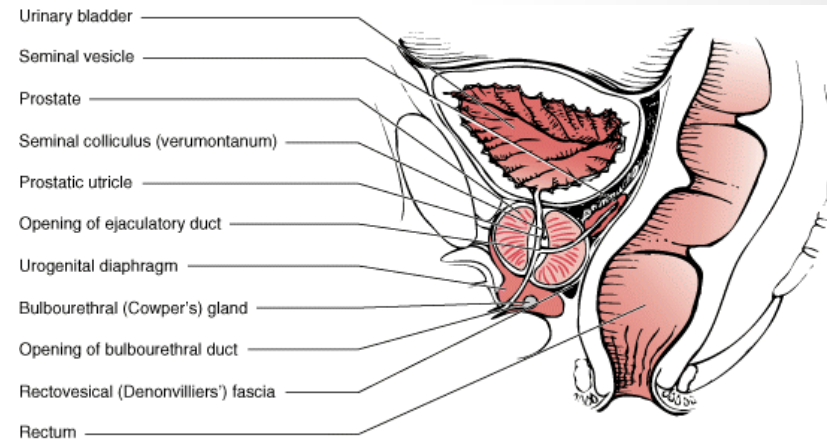
Outline

- Prostate biology – the normal prostate
- Prostate cancer epidemiology
 - Incidence and mortality in the UK
 - Screening
 - Treatment
 - Risk Factors
- Clinical outcomes for men with prostate cancer
 - Disease- related
 - Interaction with diabetes and metabolic syndrome
 - Treatment-related
 - Adverse events following androgen deprivation therapy
 - Upgrading in active surveillance

Prostate Cancer Biology

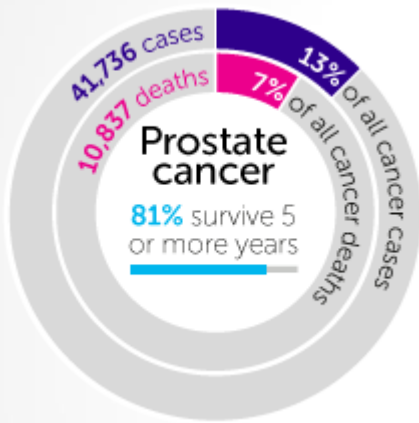
– the normal prostate

- Exocrine gland, part of the male reproductive system
- Involved in semen gelation, coagulation and liquefaction, coating and decoating of spermatozoa, and conditioning of the urethral surface
- To maintain its normal function
→ dependent on androgens, in particular testosterone



Oh W, Hurwitz M, D'Amico A, Richie J, Kantoff P. Cancer Medicine 6. London: BC Decker Inc; 2003

Prostate Cancer Epidemiology in the UK

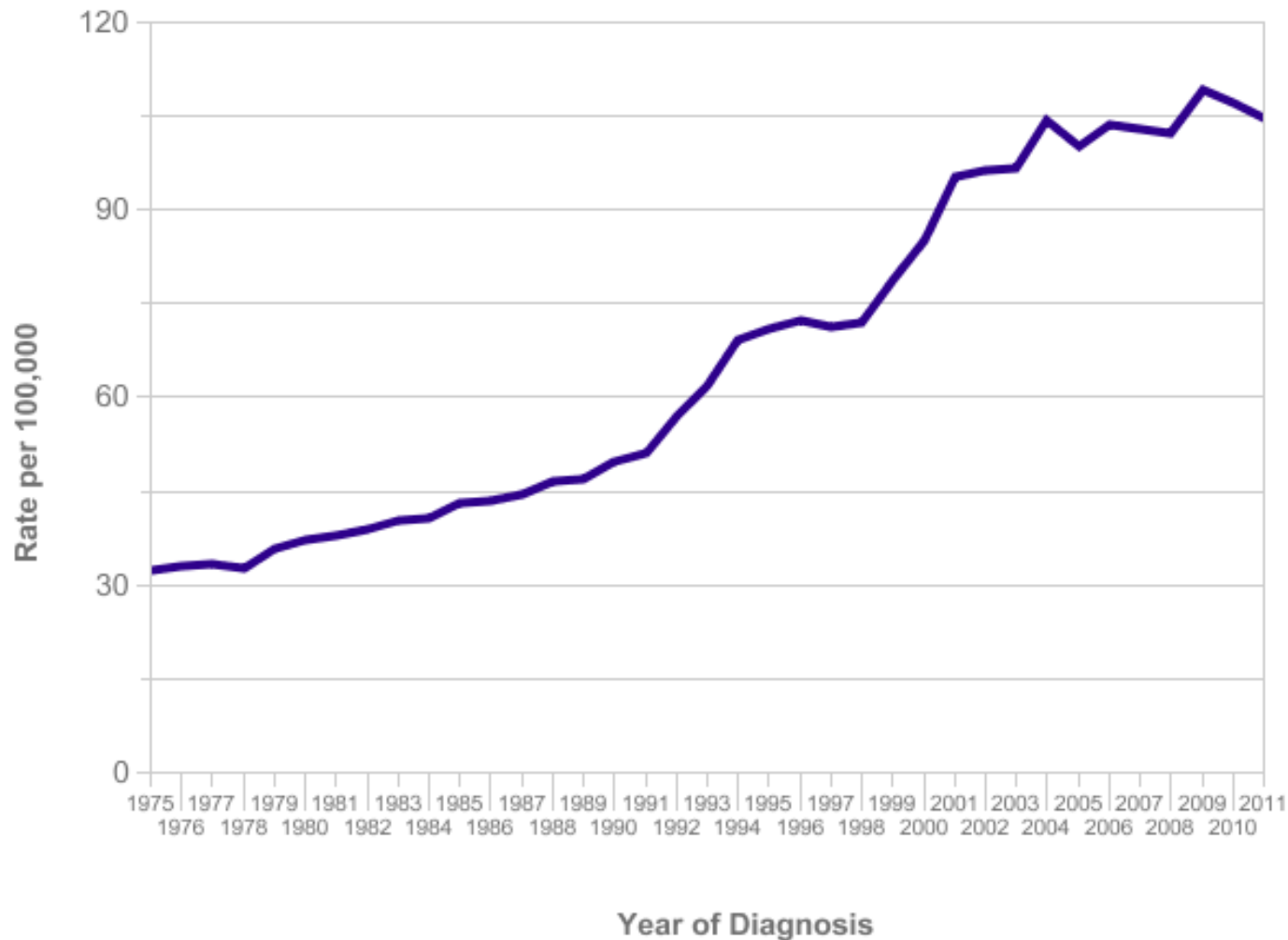


www.cruk.org

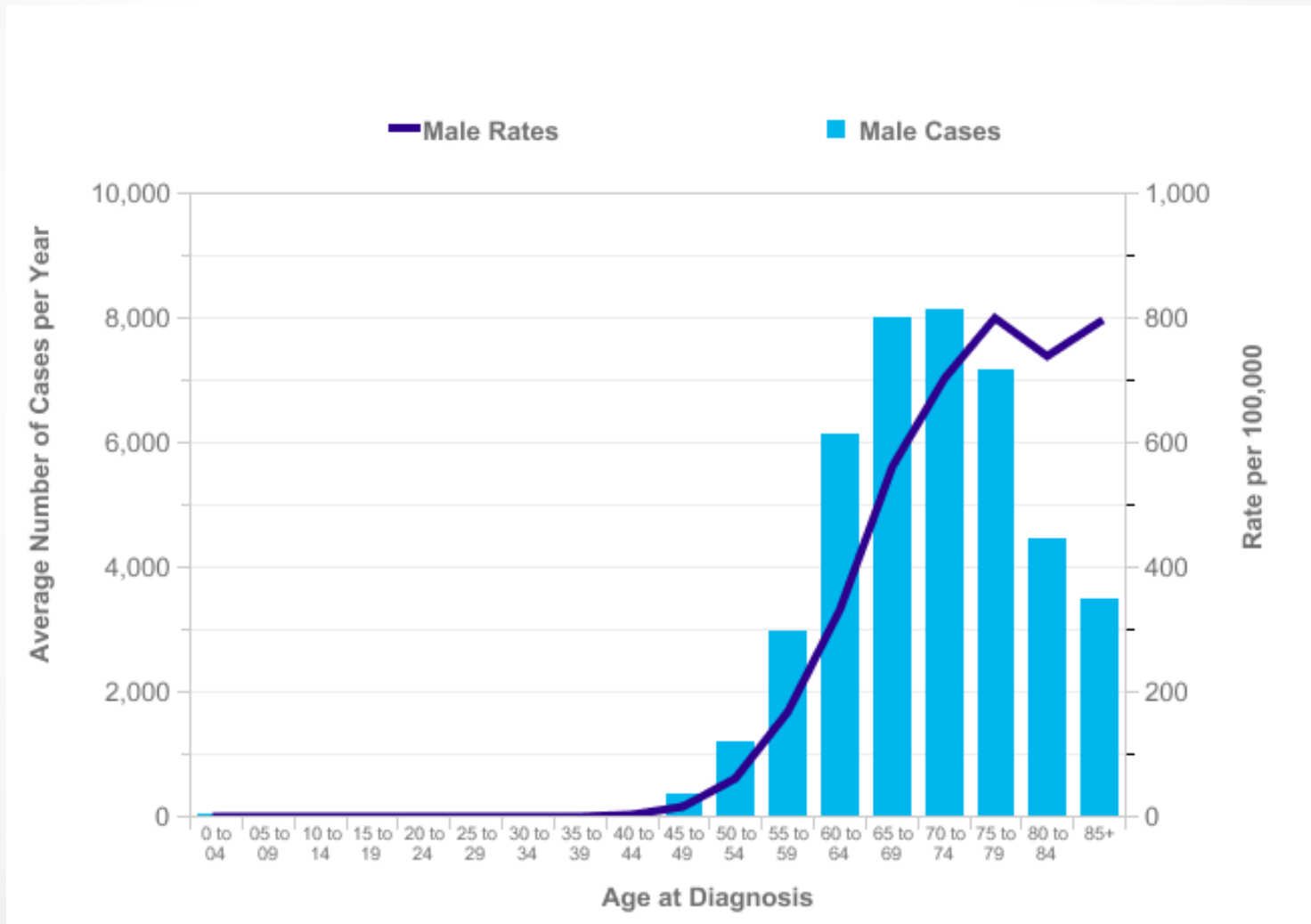
- Prostate cancer is the most common cancer in men in the UK.
- Prostate cancer accounts for a quarter of all new cases of cancer diagnosed in men in the UK.
- Over the last 35 years prostate cancer rates in Great Britain have more than tripled, although much of the increase is due to increased detection through widespread use of the PSA test.

Prostate Cancer

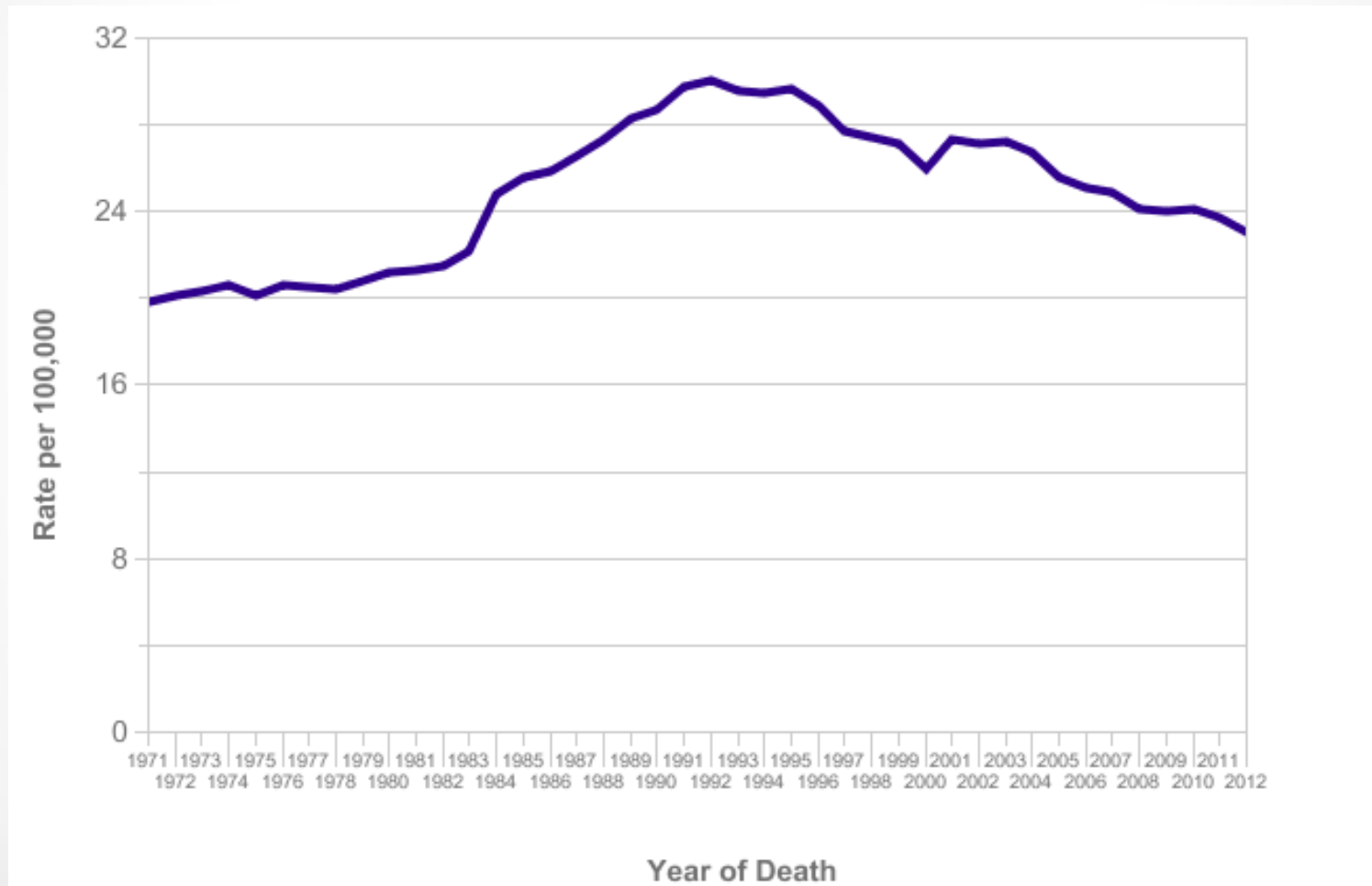
Epidemiology: Incidence



Prostate Cancer Epidemiology: Incidence by age



Prostate Cancer Epidemiology: Mortality



PSA Screening

- Appropriateness and economic cost when applied to the general public remain controversial.
- Disagreement regarding the effectiveness of PSA screening for mortality reduction and quality-of-life improvement:
 - Complex biology and long natural history of prostate cancer
 - Contradicting results of two large RCTs:
 - ERSPC
 - PLCO

ERSPC versus PLCO

- European Randomised Study of Screening for Prostate Cancer:
 - 162,000 men aged 55-69
 - After median follow-up of 0 years there was a 20% decrease in prostate cancer death, but high risk of overdiagnosis
 - 1,410 men need to be screened and 48 additional cases to be treated to prevent one death from prostate cancer
- Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial
 - After 7 years of follow-up the rate of prostate cancer death was very low, but not different between the screening group and the standard of care group
 - 44% of men had PSA tested prior to randomisation → underpowered study
- Many discussions → no conclusions yet
- UK ProtecT (Prostate testing for cancer and treatment) – results expected in 2015

Treatment

- Active Surveillance
- Watchful Waiting
- Radical Prostatectomy
- Radiotherapy
- Androgen Deprivation Therapy (ADT)
- Chemotherapy

Risk Factors

- Age
- Family History:
 - First-degree relative → 2 to 3-fold increased risk of prostate cancer
- Ethnicity
 - Black African-Caribbean men → 3-fold increased risk of prostate cancer
- Metabolic syndrome – lipid metabolism?

Metabolic syndrome

- Defined by a combination of metabolic risks:

Metabolic syndrome
(Syndrome X)

- Central obesity
- High blood pressure
- High triglycerides
- Low HDL-cholesterol
- Insulin resistance



<http://www.americanhealthjournal.com>

- Three of the above listed metabolic risks = MetS¹

¹Alberti K et al. Circulation 2009.

Epidemiology of MetS

- In the US: 34.6% of the population meets the NCEP criteria for the metabolic syndrome, (NHANES, 1999-2002).
- UK – data from the national Health Survey

Component	Men	Women
Overweight/Obese	67%	56%
Diabetes	5.6%	4.2%
HDL <1 mmol/L	9.4%	1.8%
Hypertension	31%	28%
Triglycerides ≥ 1.6 mmol/l	41.2%	30.8%

Metabolic syndrome and cancer

- Migrant studies: “long term immigrants” approach cancer risk of men of the country they migrate to
 - Life style changes affect cancer risk?
 - Metabolic syndrome reflects part of people's lifestyle
- Most studies focus on the link between obesity or diabetes and cancer risk
- Metabolic syndrome is found to increase the incidence of colorectal, prostate, and breast cancer
- Link is of public health importance as metabolic syndrome can be prevented

Obesity and Cancer

- Estimated 17,294 excess in cancer cases occurring in 2010 due to overweight and obesity (5.5% of all cancers).
- Recommendation from the World Cancer Research Fund

RECOMMENDATION 1

BODY FATNESS

Be as lean as possible within the normal range¹ of body weight

PUBLIC HEALTH GOALS

Median adult body mass index (BMI) to be between 21 and 23, depending on the normal range for different populations²

The proportion of the population that is overweight or obese to be no more than the current level, or preferably lower, in 10 years

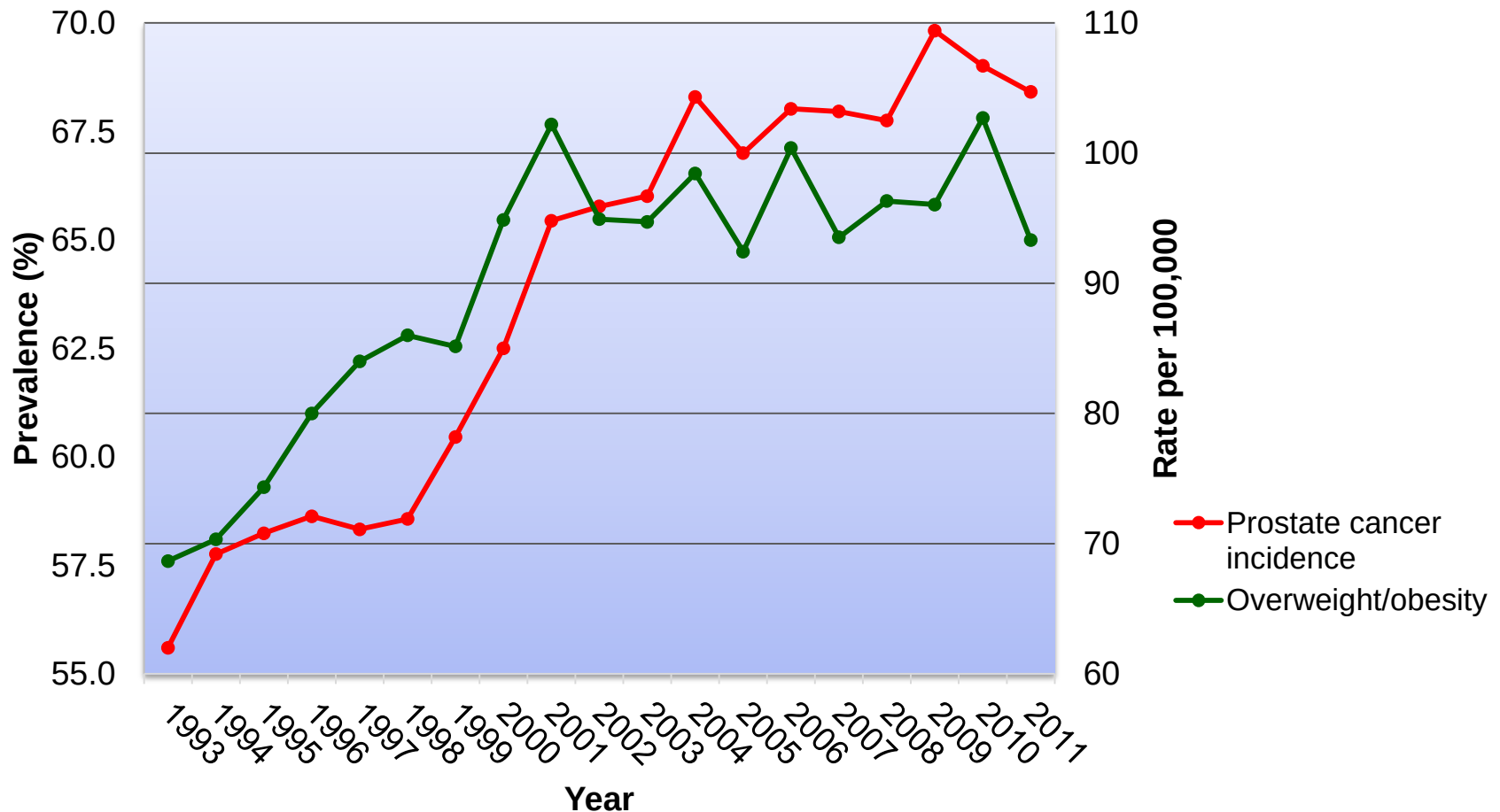
PERSONAL RECOMMENDATIONS

Ensure that body weight through childhood and adolescent growth projects³ towards the lower end of the normal BMI range at age 21

Maintain body weight within the normal range from age 21

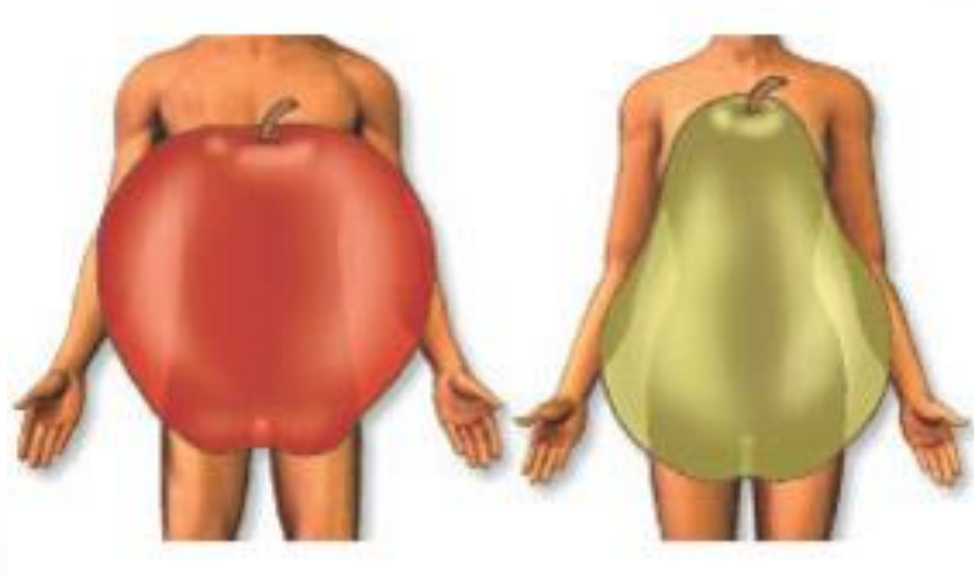
Avoid weight gain and increases in waist circumference throughout adulthood

Overweight/obesity and prostate cancer in England



Underlying mechanisms

- Plausible link based on different studies
- Pathways remain hypothetical but mainly focus on diabetes and obesity:
 - Insulin and IGF-I activities
 - Adiponectin
 - Fatty acids
 - Epigenetics



Prostate Cancer Risk in the Swedish AMORIS Study

The Interplay Among Triglycerides, Total Cholesterol, and Glucose

Mieke Van Hemelrijck, PhD¹; Hans Garmo, PhD^{1,2}; Lars Holmberg, PhD^{1,2}; Göran Walldius, PhD^{3,4}; Ingmar Jungner, PhD⁵; Niklas Hammar, PhD^{4,6}; and Mats Lambe, PhD^{2,7}

Low levels of apolipoprotein A-I and HDL are associated with risk of prostate cancer in the Swedish AMORIS study

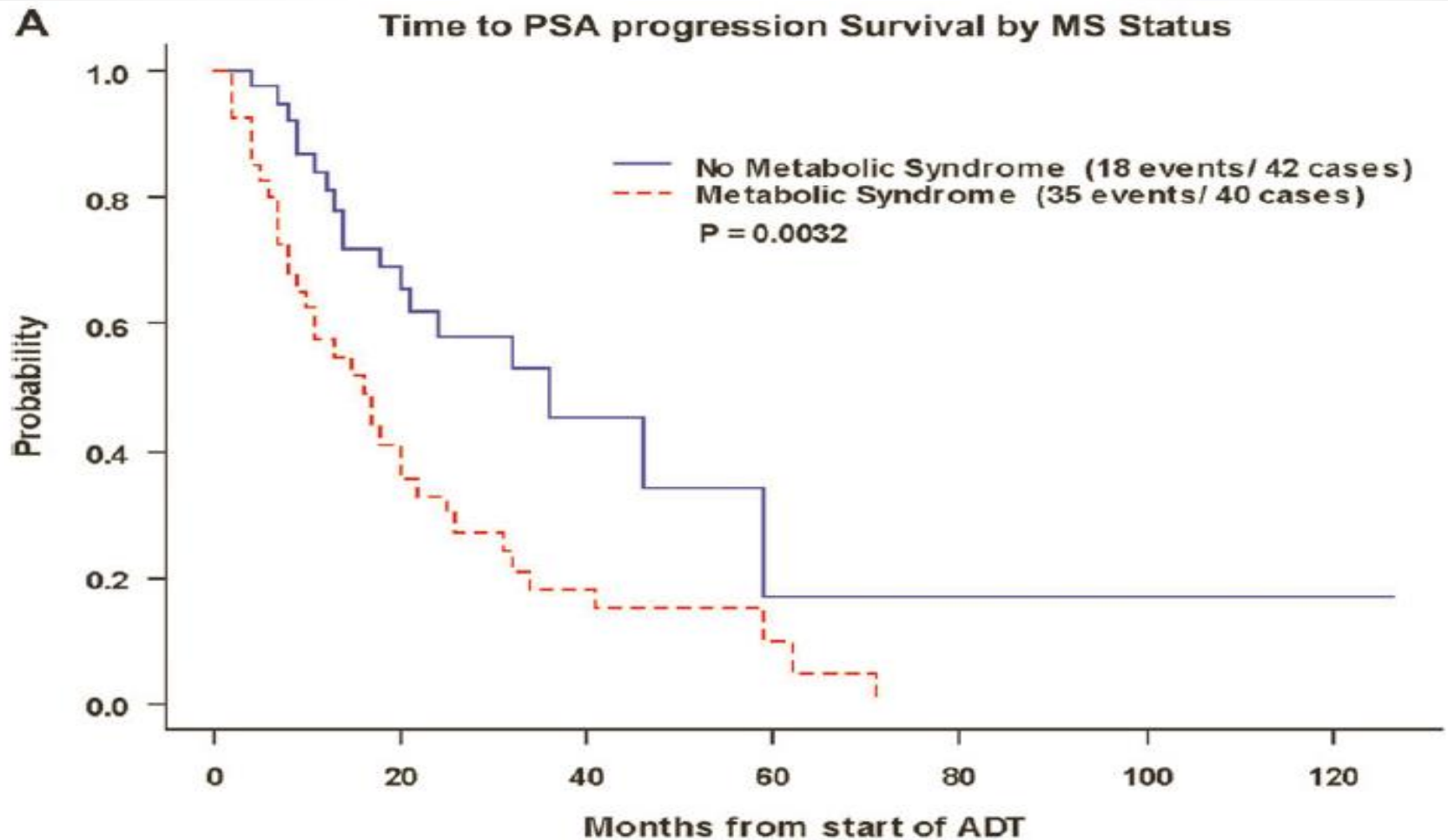
Mieke Van Hemelrijck • Göran Walldius • Ingmar Jungner •
Niklas Hammar • Hans Garmo • Elisa Binda •
Adrian Hayday • Mats Lambe • Lars Holmberg

Risk of prostate cancer is not associated with levels of C-reactive protein and other commonly used markers of inflammation

Mieke Van Hemelrijck¹, Ingmar Jungner², Göran Walldius^{3,4}, Hans Garmo^{1,5}, Elisa Binda⁶, Adrian Hayday^{6,7}, Mats Lambe^{5,8}, Lars Holmberg^{1,5} and Niklas Hammar^{3,9}

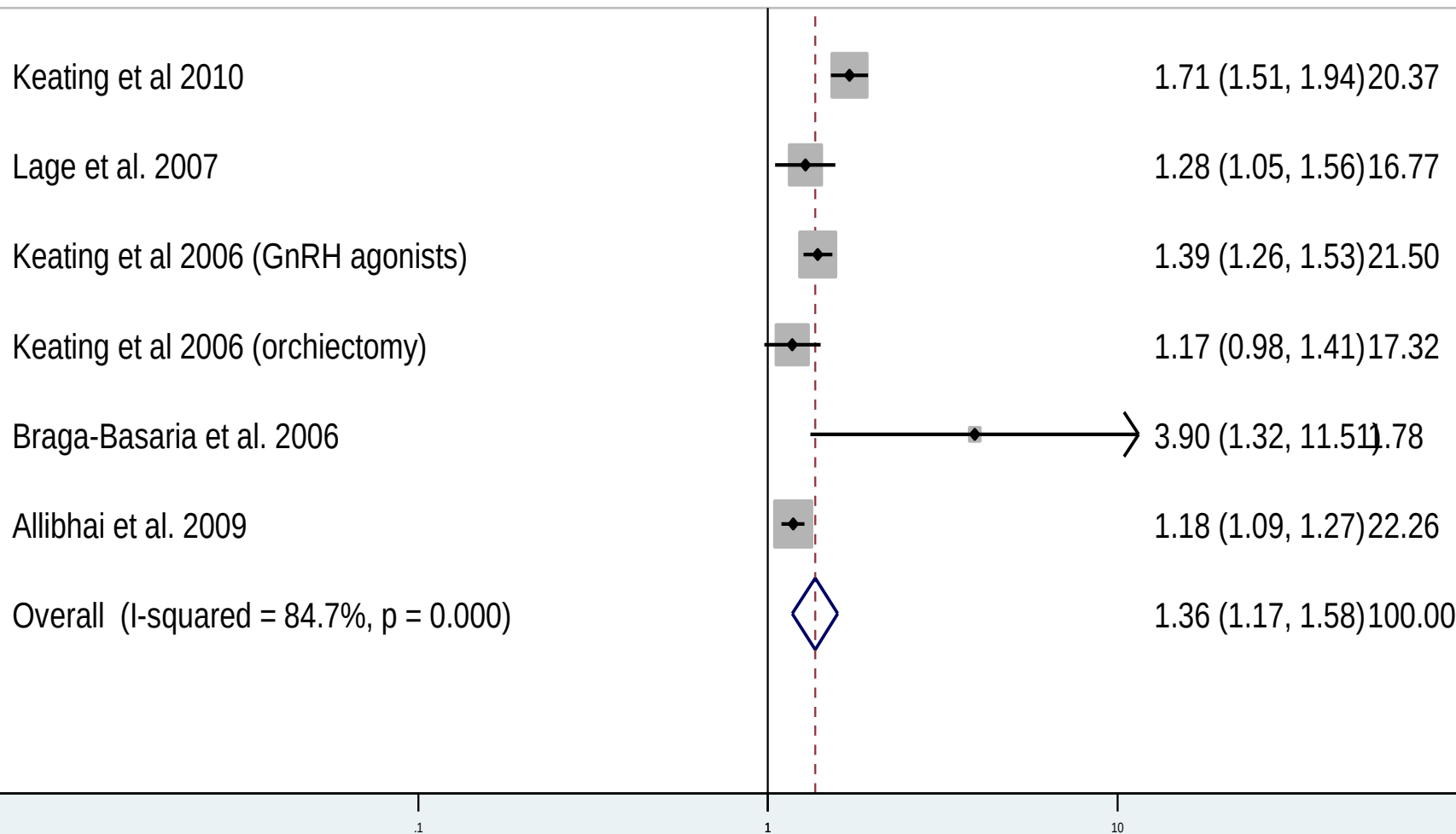
MetS and response to Androgen Deprivation Therapy (ADT)

Flanagan et al. Ann Oncol 2011.



Androgen deprivation therapy and risk of diabetes

Bosco et al. Unpublished



Statins and PCa death

- Reduce levels of cholesterol in the blood
- Observational studies suggest statins reduce cancer progression
- Well tolerated, with minimal and rare side effects
- No RCTs available yet
- Use of causal inference methods to treat observational data as if they came from an RCT
 - Effect of statins on risk of PCa-death cannot be reliably discerned.
 - Data on serum cholesterol and disease severity in one setting are needed
 - **There is a need for a well-defined clinical trial investigating the effect of statins on cancer-specific death**

Androgen Deprivation Therapy

Prostate Cancer (PCa)

- Most common cancer for men.
- 1/4 of newly diagnosed male cancers.



Androgen deprivation therapy (ADT)

- Standard treatment regimen for locally advanced or metastatic disease
- Types: orchiectomy, anti-androgens (AA), gonadotropin releasing-hormone (GnRH) agonists/antagonists and combined androgen blockade (CAB)

Objective: To assess cardiovascular side-effects following ADT

Side-effects

- Metabolic changes: weight gain, hot flushes, insulin resistance, etc
- **Cardiovascular disease (CVD)**

VOLUME 28 · NUMBER 21 · JULY 20 2010

JOURNAL OF CLINICAL ONCOLOGY

ORIGINAL REPORT

Absolute and Relative Risk of Cardiovascular Disease in Men With Prostate Cancer: Results From the Population-Based PCBaSe Sweden

Mieke Van Hemelrijck, Hans Garmo, Lars Holmberg, Erik Ingelsson, Ola Bratt, Anna Bill-Axelsson, Mats Lambe, Pär Stattin, and Jan Adolfsson

Results

- Increased risks for incidence & mortality of heart disease:

Total ET group	SIR	95% CI	Obs/Exp	SMR	95% CI	Obs/Exp
Myocardial Infarction	1.25	(1.20- 1.31)	2080 / 1660.2	1.25	(1.18 - 1.32)	1191/954
Arrhythmia	1.20	(1.14 - 1.26)	1692 / 1410.8	1.01	(0.87 - 1.16)	188/187
Ischaemic Heart Disease	1.32	(1.27 - 1.36)	3284 / 2492.6	1.21	(1.16 - 1.26)	2097/1736.4
Heart failure	1.32	(1.27 - 1.38)	2140 / 1619.3	1.24	(1.12 - 1.37)	372/299.5
Stroke	1.26	(1.21 - 1.30)	2619/2085.5	1.24	(1.14 - 1.35)	546/441.5

•SIRs and SMRs > 1

•Increased risks

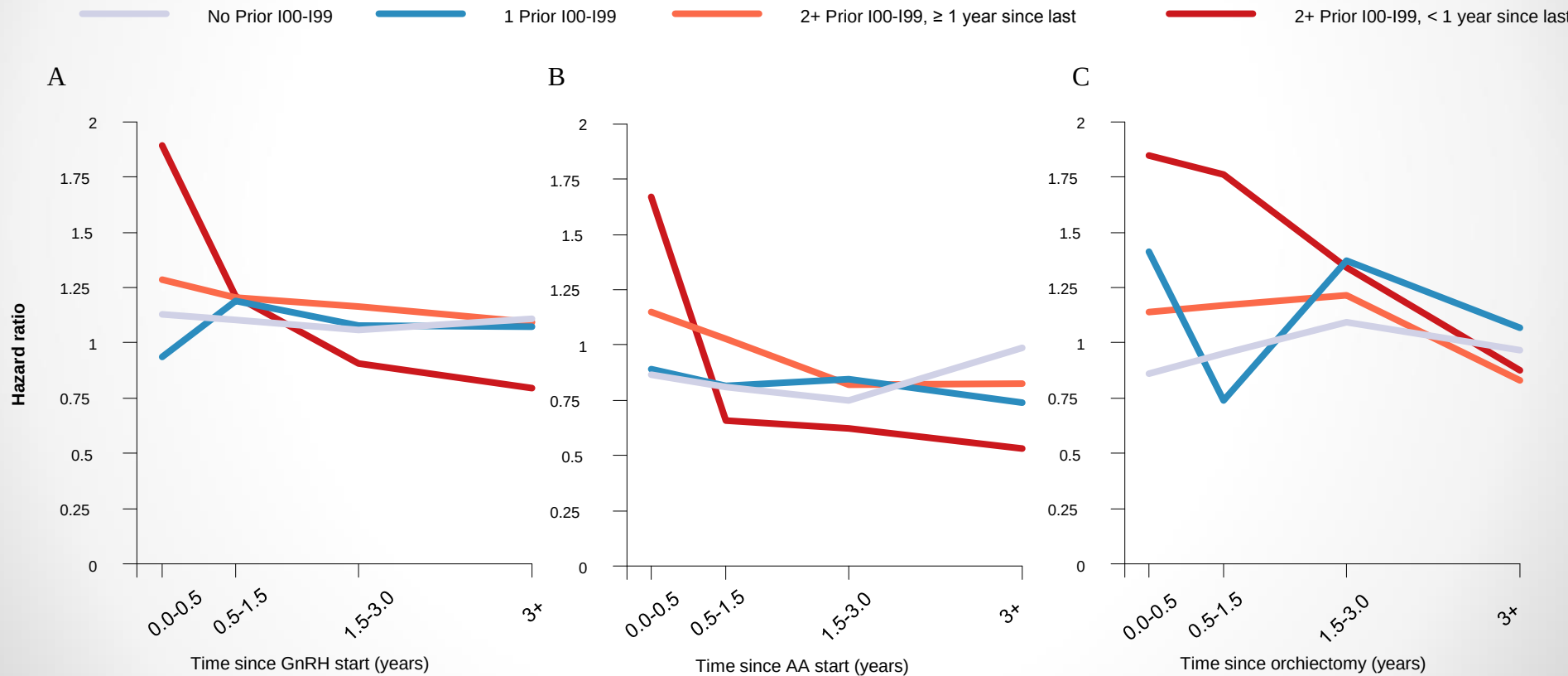
Main findings

- Complex pattern of increased CVD risk
- Risk of nonfatal and fatal CVD was higher for those on ADT, with generally lower risk for those treated with anti-androgens than those treated with other types of ADT
- Largest absolute risk difference for nonfatal CVD was among those on ADT.
- Absolute risks were higher for men with a history of CVD compared to those without, despite the more modest relative risks in this group.

Summary: ADT and CVD

- Important to take heart disease risk into account when prescribing ADT for prostate cancer
- Need for studies verifying this association and exploring plausible biological mechanisms
 - Would help to tease out when to best use these treatments according to a patient's history of heart disease
 - Would help to identify whether it is a good idea to give patients heart medicines to counteract these side effects

Risk of CVD since ADT initiation



Conclusion: ADT and CVD

- Although many steps remain to understand if ADT alone causes the increased risk or if it acts together with other factors, our findings highlight problems that require further investigation for a common clinical situation in oncology.
- The potential value of applying suitable lifestyle changes or cardio protective therapy should be investigated in men who present with baseline CVD and it could be contemplated to be observant of signs of CVD during the first year of treatment.

Risk of thromboembolic diseases in men with prostate cancer: results from the population-based PCBaSe Sweden

Mieke Van Hemelrijck, Jan Adolfsson, Hans Garmo, Anna Bill-Axelsson, Ola Bratt, Erik Ingelsson, Mats Lambe, Pär Stattin, Lars Holmberg

ADT and TED

- Important to consider TED when treating men with PCa, especially those on ADT
- Higher risk of TED was noted for the younger men and men with metastatic disease
- Risk patterns of TED are probably explained by PCa, treatment, and factors taken into consideration when selecting treatments (i.e. age, tumour stage).

available at www.sciencedirect.com
journal homepage: www.europeanurology.com



European Association of Urology



Platinum Priority – Prostate Cancer

Editorial by Matthew R. Cooperberg on pp. 701–703 of this issue

Multiple Events of Fractures and Cardiovascular and Thromboembolic Disease Following Prostate Cancer Diagnosis: Results From the Population-Based PCBaSe Sweden

Mieke Van Hemelrijck^{a,}, Hans Garmo^{a,b}, Lars Holmberg^{a,b}, Pär Stattin^c, Jan Adolfsson^d*

ADT and multiple adverse events

- Clinical information routinely available for PCa patients cannot identify men at increased risk of multiple adverse events with enough precision to be of help in designing surveillance programs.
- Primary events of HD or stroke after PCa Diagnosis were associated with an increased risk of multiple events of HD or stroke.

Active Surveillance versus Watchful Waiting

- Localised prostate cancer
- **Watchful waiting** ~ 'symptom-guided treatment':
 - Generally well accepted for men with low-risk disease or men with a life expectancy of fewer than 10 years.
 - Despite its advantage of avoiding potential morbidities from radical local therapies, a large proportion of these men still needs palliative therapy
- **Active surveillance:**
 - When the risks of radical treatment outweighs the benefits, but this can change in the light of new information (i.e. short PSA doubling time and deteriorating histopathological factors on repeat biopsy).
 - Continues as long as that remains, but ends with either treatment, death from another cause or transition to watchful waiting

Active Surveillance

- Management option for men with low risk prostate cancer
- Estimated that men remain on AS between 2.2 and 5.4 years with an estimated probability of discontinuation of 33% at 5 years and of 55% at 10 years
- Discontinuation occurs due to either disease progression or patient anxiety
 - Disease progression can potentially be avoided if high-risk disease is adequately assessed at the outset
 - Anxiety can be reduced by providing PCa men with more confidence in the low risk nature of their disease
- Main advantage: low treatment morbidity, but existing nomograms, based on transrectal prostate biopsy (TRBx), have only modest accuracy in predicting the outcomes of prostate cancer men managed by AS

Confirmatory transperineal sector prostate biopsy for assessment of prostate cancer

- We aimed to compare the histopathological results of TPSBx samples with those of TRBx from men considered suitable for AS following a TRBx-derived PCa diagnosis.
- ➔ AS selection defined by TRBx misclassified more than half of men irrespective of disease severity at baseline. More than 30% of men with apparent low risk disease on TRBx were found to have higher-risk disease according to the results from systematic TPSBx.

What does a second transperineal sector prostate biopsy offer for men on active surveillance?

- This single-centre study suggests that histological progression occurs in approximately one third of men with prostate cancer on AS, despite a favourable initial TRBx and confirmatory TPSBx. A second TPSBx at 2 years is essential to exclude histological progression in those who might benefit from active intervention.

Questions?